EXTERN + OUTAMATION
EXTERNSHIP

ADVANCED AI-POWERED

DOCUMENT INSIGHT AND DATA EXTRACTION PROJECT

SPRINT 8 STATUS UPDATE

by Lashawn Fofung, MBA, PMP, CSM, CSPO

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SKILLS

- Python
- AI/ML
- Data Scrapping
- Prompt Engineering
- LLMs
- Text Extraction
- Document Intelligence
- Python Automation
- Optical Character Recognition (OCR)
- Retrieval-Augumented Generation (RAG)

Helping businesses improve, automate, and architect critical tasks (such as document handling, customer service, and data management) by saving thousands of hours and boosting accuracy.

Mortgage Documents

SPRINTS

2025

01 AI Document Intelligence

02 Python & Google Colab

03 Python Data Extraction

04 Optimizing OCR

05 Implementing RAG

06 Optimizing RAG Pipelines

07 Blob Processing & Classification

08 Interactive Chatbot

09 AI-Powered Document Automation Platform

PROJECT TIMELINE

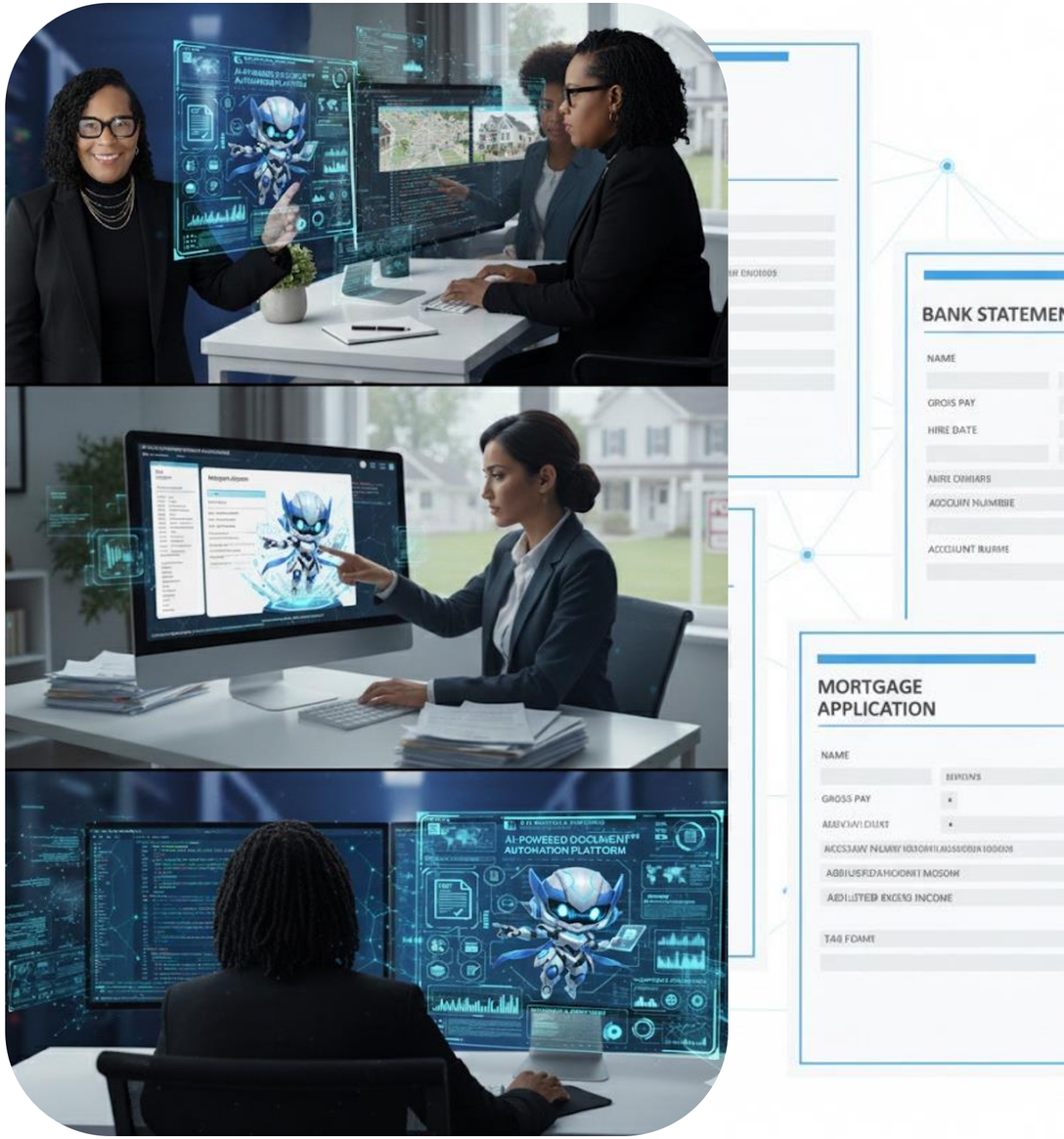
10 WEEKS

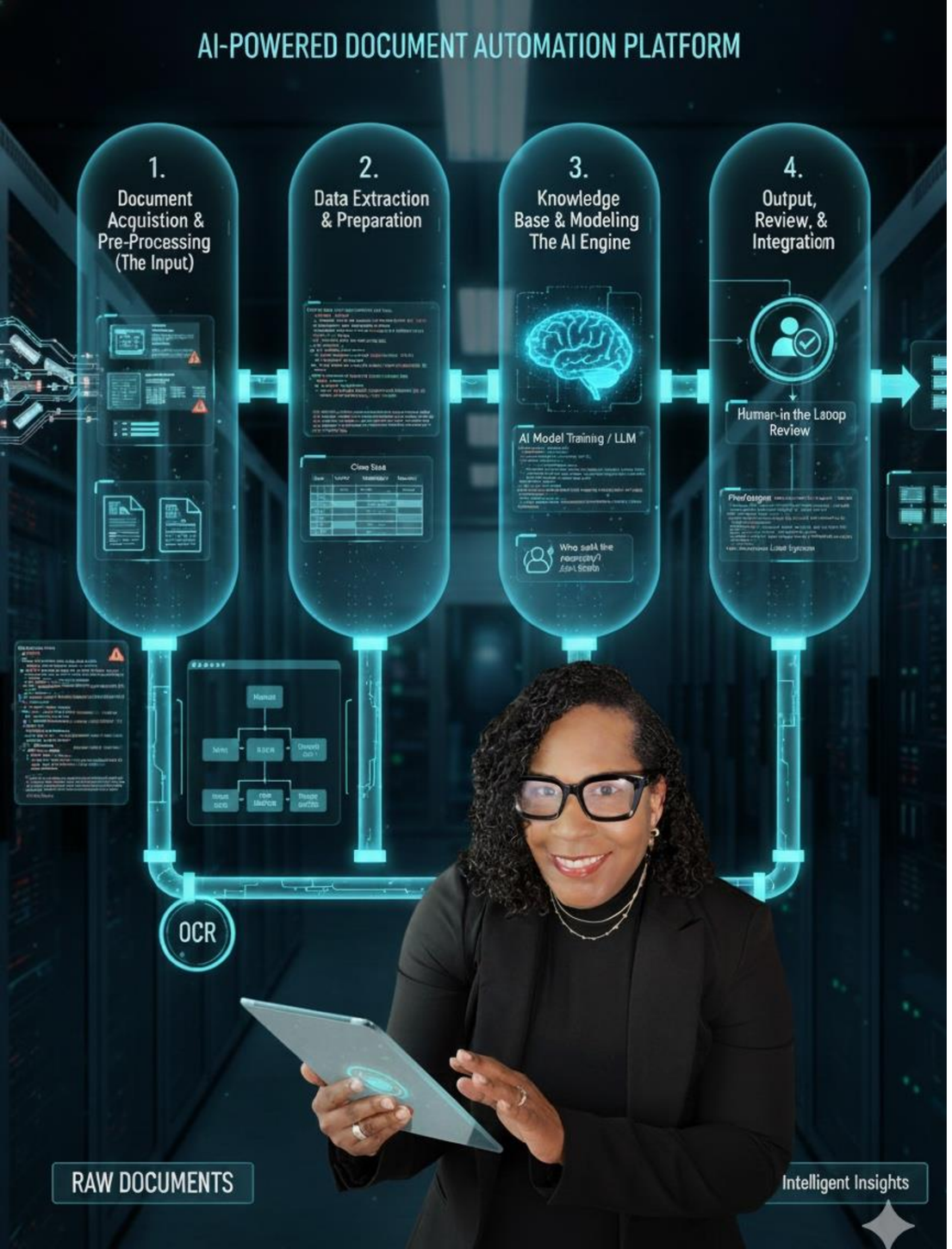
Interactive Chatbot

Gradio Integration and RAG

Background: 10-Week Project

The Outamation Advanced AI-Powered Document Insights and Data Extraction Externship is a comprehensive 10-week program. Its core goal and purpose are to master the required skills to create an AI-Powered Document Automation Platform. The program focuses on bridging foundational technologies (ML, LLMs, NLP, Computer Vision) with high-value industry applications, such as solving complex, document-heavy challenges in mortgage automation, ultimately transforming slow, manual review processes into automated, efficient workflows.





Challenge & Solution: Building The Document Intelligence (AI) Pipeline

- **The Business Challenge:**
  - Automating the highly manual and time-consuming process of mortgage underwriting by handling 200-300 pages of unstructured documents.

- **The Solution:**
  - Designing and implementing a robust, AI-Powered Document Automation Platform to transform raw files into structured, intelligent insights.

- **The 4-Step Document Intelligence Workflow:**
  - Step 1. Document Acquisition & Pre-Processing (The Input)
  - Step 2. Data Extraction & Preparation
  - Step 3. Knowledge Base & Modeling (The AI Engine)
  - Step 4. Output, Review, & Integration (The Results)

SPRINT 8 FOCUS

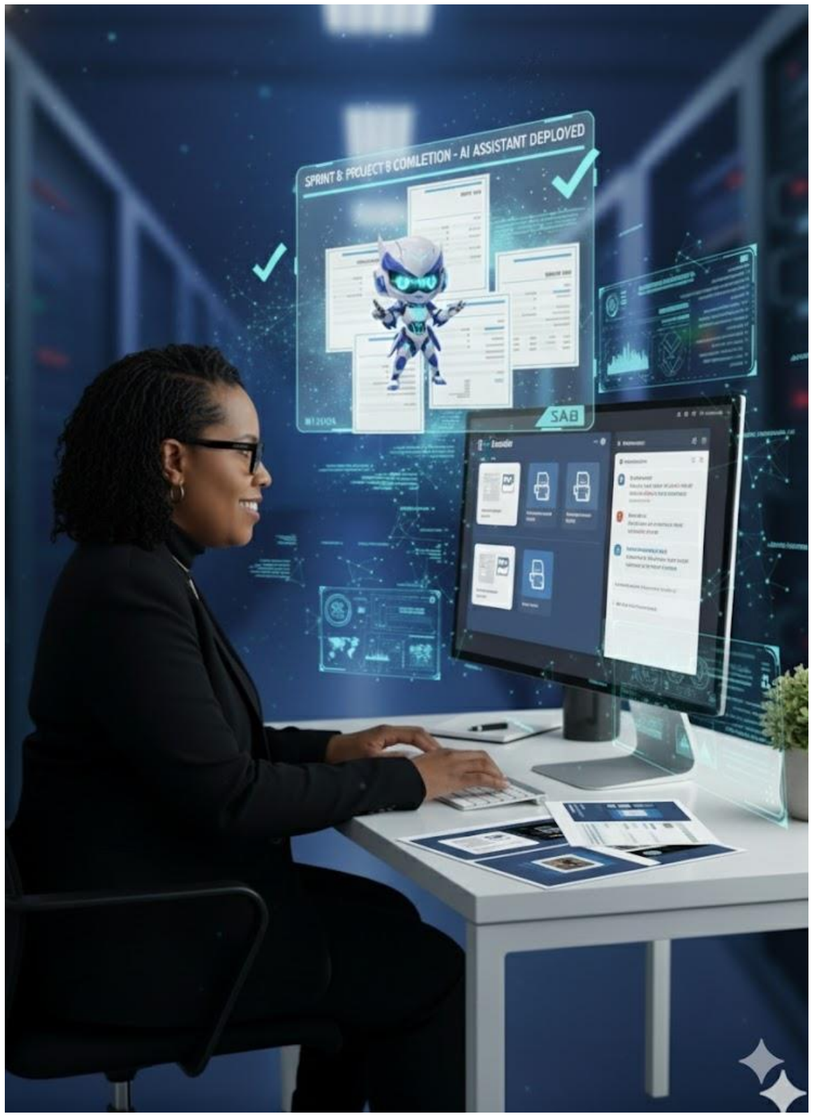
Step 4

Sprint 8 focuses on bridging the gap between a backend RAG (Retrieval-Augmented Generation) pipeline and a user-facing AI assistant. This sprint transitions the work from a developer-centric notebook environment to a functional product designed for non-technical users.

Overall Status: Sprint 8 Complete

- The primary goal was to bridge the gap between a technical backend RAG (Retrieval-Augmented Generation) pipeline and a user-facing AI assistant.

- The project transitioned the system from a developer-centric notebook to an interactive, web-based chatbot designed for non-technical users.



Key Highlights

- **Modularization:** Shifting from procedural code to reusable functions to ensure the pipeline is "UI-ready" and can integrate with various LLMs (e.g., Gemini, Mistral, or Phi-2).

- **User Experience (UX):** Moving beyond command-line queries to allow users to interact with documents through a clean browser interface similar to ChatGPT.

- **Real-World Application:** Solving the "Steve from Loan Processing" problem which is enabling team members without coding knowledge to use advanced AI tools.

Key Technical Components

- **Frontend Framework:** Gradio is used to build the interface without requiring HTML or JavaScript knowledge.
- **LLM Orchestration:** Integration with Google Gemini 2.0 Flash for reasoning and generating responses.
- **Retrieval Engine:** Implementation of a full RAG pipeline using LlamaIndex for parsing, chunking, and indexing PDFs.
- **Semantic Search:** Utilizing the BGE (BAAI/bge-small-en-v1.5) model via Sentence-Transformers to understand query meaning rather than just keyword matching.
- **Document Intelligence:** Using AI to identify document boundaries (e.g., distinguishing between a Resume and a PaySlip within a single PDF).

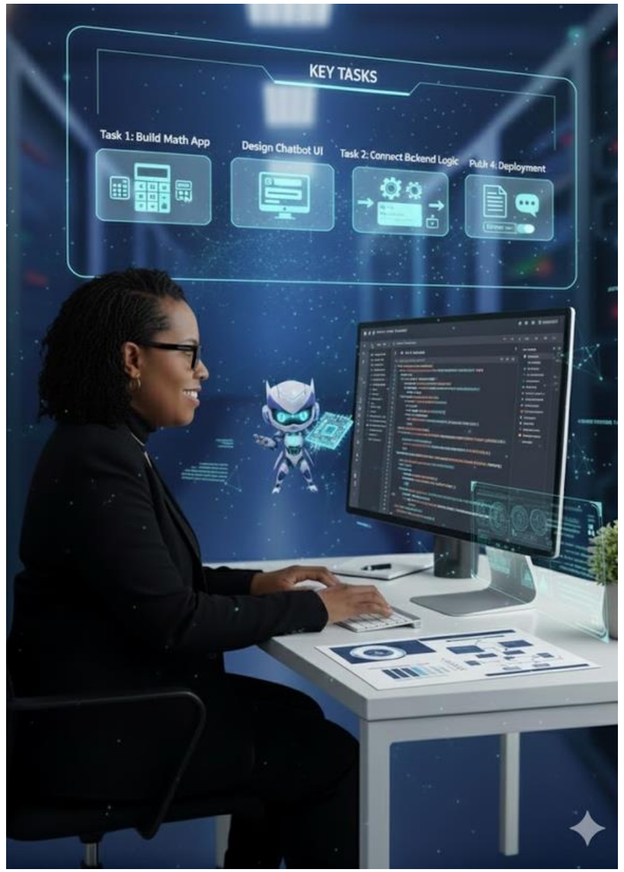
Key Tasks

- **Build a Simple AI Assistant:** A preliminary task to master Gradio basics AI assistant to answer questions about document.

- **Design the Chatbot UI:** Setting up the gr.Blocks layout, including chat windows, text boxes, and file upload components.

- **Connect Backend Logic:** Replacing "mock" response functions with a dynamic RAG pipeline that actually reads and queries uploaded PDFs.

- **Public Deployment:** Using Gradio’s share=True feature to generate a live link for remote testing.



Key Deliverables

- **Functional Web App:** A Gradio-powered application with PDF upload capabilities and a persistent chat history.
- **Modular Codebase:** A clean Python project with separated logic for the UI and the backend retrieval engine.
- **Presentation (PPT Deck):** A 3–4 slide deck featuring screenshots of the UI in action and a reflection on changes made from the basic tutorial.
- **Custom Features:** At least one unique enhancement, such as image/audio support, multiple file uploads, or a "loading" status indicator.

CoLab Notebook

- **Intro to Gradio**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/Gradio/Task_Intro_to_Gradio.ipynb

- **Gradio Chatbot with Lite RAG Implementation**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/Gradio/Task_Intro_to_Gradio.ipynb

- **Full RAG Pipeline with Interactive Gradio Chatbot**
  - https://github.com/LashawnFofung/RAG-Pipelines/blob/main/Gradio/Task_Full_RAG_Pipeline_with_Interactive_Gradio_Chatbot.ipynb

THANK YOU

https://github.com/LashawnFofung/AI-Powered-Document-Automation-Platform